CLASS 10 — SCIENCE FULL SYLLABUS PAPER SET 07

Full Syllabus (Chapters 1–13) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The thermite reaction used for joining railway tracks involves: 1

(a) Fe_2O_3 and aluminium **(b)** zinc and iron
(c) aluminium and copper **(d)** iron and copper

2. The gas evolved when an acid reacts with a metal hydrogencarbonate is: 1

(a) Cl_2 **(b)** O_2 **(c)** H_2 **(d)** CO_2

3. Common salt is obtained from sea water by: 1

(a) evaporation **(b)** sublimation **(c)** filtration **(d)** distillation

4. The impurities present in an ore are called: 1

(a) slag **(b)** anode mud **(c)** flux **(d)** gangue

5. Student A says zinc oxide is amphoteric; Student B says it is acidic only. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. Soaps are sodium or potassium salts of: 1

(a) short-chain alcohols **(b)** long-chain carboxylic acids
(c) mineral acids **(d)** sulphonic acids

7. Which of these will give a sooty flame? 1

(a) LPG **(b)** naphthalene **(c)** ethanol **(d)** methane

8. Glomerulus is a cluster of: 1

(a) blood capillaries **(b)** muscle cells
(c) nerve cells **(d)** lymph vessels

9. Oestrogen is secreted by: 1

(a) pancreas **(b)** ovaries **(c)** testes **(d)** thyroid

10. The number of chromosomes in a human zygote is: 1

(a) 22 **(b)** 46 **(c)** 23 **(d)** 92

11. Fertilisation results in the formation of: 1

(a) bud **(b)** gamete **(c)** zygote **(d)** spore

12. A student states: “Each human gamete contains one sex chromosome.” The correct response is: 1

(a) Correct **(b)** Only in males
(c) Incorrect **(d)** Cannot be determined

13. Mendel studied how many pairs of contrasting characters in pea? 1

(a) two **(b)** five **(c)** nine **(d)** seven

14. The speed of light in vacuum is: 1

(a) 3×10^6 m/s **(b)** 3×10^8 m/s **(c)** 3×10^{10} m/s **(d)** 3×10^5 m/s

15. The phenomenon of light involved in the formation of a rainbow includes: 1

(a) refraction, dispersion and internal reflection
(b) only reflection
(c) diffraction only
(d) only scattering

16. Energy consumed by a 500 W heater in 4 hours is: 1

(a) 20 kWh **(b)** 2 kWh **(c)** 2000 kWh **(d)** 0.2 kWh

17. 1 ampere is equal to: 1

(a) 1 C/s **(b)** 1 V/s **(c)** 1 J/C **(d)** 1 J/s

18. An electric fuse works on the: 1

(a) heating effect of current **(b)** chemical effect
(c) magnetic effect **(d)** photoelectric effect

19. In an ecosystem, energy is lost at each trophic level mainly as: 1

(a) light **(b)** electricity **(c)** sound **(d)** heat

20. Rainfall, temperature and soil are: 1

(a) biotic components **(b)** abiotic components
(c) consumers **(d)** producers

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Why is hydrogen peroxide stored in dark coloured bottles? 2

22. Name the acid present in 2
(i) tamarind
(ii) orange.

23. Why does the rate of breathing in aquatic organisms differ from terrestrial ones? 2

24. A student says that the cerebellum controls hunger. Justify or refute. 2

25. What is the role of the cotyledons in a seed? 2

26. State the law of segregation. 2

27. Why is a convex mirror preferred as a rear-view mirror? 2

28. What is the function of an MCB? 2

MASTERMIND

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Why is the reaction of an acid with a base called double displacement? Write the equation for NaOH and HCl and state the ions exchanged. 3

30. Describe an activity to show that metals are good conductors of electricity. 3

31. Explain the feedback mechanism with reference to blood sugar. 3

32. What are the functions of the 3
(i) fallopian tube
(ii) uterus
(iii) placenta?

33. The near point of a hypermetropic eye is 1 m. Find the power of the lens needed to correct it (normal near point 25 cm). 3

34. What is meant by electric fuse? Explain its working in a domestic circuit. 3

35. Explain how acidic and basic solutions behave with 3
(i) blue litmus
(ii) red litmus
(iii) methyl orange.

OR

What is a neutralisation reaction? Give an equation, and one example of neutralisation in daily life. 3

36. Explain why the impact of removing all organisms in a trophic level differs for different levels. 3

OR

What is biological magnification? Why is the concentration of DDT highest in humans? 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) Explain the New Cartesian sign convention for lenses. 5
(ii) A concave lens has focal length 20 cm. At what distance should an object be placed so that it forms an image at 15 cm?

OR

(i) State the New Cartesian sign convention for spherical mirrors. 5
(ii) Find the focal length of a convex mirror whose radius of curvature is 32 cm.
(iii) Why is it used in vehicles?

38. **Case study:** Compound A ($C_2H_4O_2$) gives brisk effervescence with sodium carbonate. A reacts with B (C_2H_6O) in the presence of conc. H_2SO_4 to give C with a fruity smell. 5
(i) Identify A, B and C.
(ii) Write the equation for A with Na_2CO_3.
(iii) Name the reaction of A with B.
(iv) What happens when C is heated with NaOH?

OR

(i) Explain the oxidation of ethanol with an equation. 5
(ii) What are oxidising agents? Give two examples.
(iii) Distinguish combustion from oxidation.

39. **Case study:** In a house, a 2 kW geyser, three 100 W fans and five 20 W LED bulbs run on a 220 V supply. **5**

(i) Find the total power.

(ii) Find the current drawn.

(iii) Which fuse rating is suitable: 5 A or 15 A?

(iv) Energy used if all run for 3 h.

(v) Why are they connected in parallel?

OR

Case study: A student connected two identical bulbs first in series and then in parallel to the same battery. **5**

(i) In which case is total resistance more?

(ii) In which case do the bulbs glow brighter?

(iii) If each bulb has 10 Ω and the battery is 10 V, find the current in each case.

(iv) What happens if one bulb fuses in each case?

40. (i) Draw a schematic diagram of blood flow (describe) in the human heart and body. **5**

(ii) What is meant by systole and diastole?

OR

(i) Describe the process of digestion in the small intestine. **5**

(ii) What are villi? State their function.

(iii) Name the enzymes of pancreatic juice.

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND